James Nithil V

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EXPERIENCE

Junior Software Developer **Aug 2025 to Present**

Vaken Technologies Pvt Ltd | Product: Sovablu *Trichy, India*

- Engineered and scaled enterprise RESTful APIs using **Java 21, Spring Boot, and PostgreSQL**, implementing strict **Spring Transaction management and ACID principles** across critical mutation workflows to maintain data consistency and prevent data loss under concurrent client traffic.
- Architected a hybrid persistence layer pairing **PostgreSQL** with **Amazon DynamoDB** for fast key value reads. Restructured table designs and eliminated redundant columns to reduce I/O payload size while implementing **composite indexing and query optimization** to eliminate full row scans.
- Implemented an in memory **Redis caching tier with TTL expiration** for frequently queried datasets to reduce repeated database queries. Applied **cache stampede prevention mechanisms** using mutex locks to safeguard the database from concurrent traffic spikes and improve API response times.
- Secured file and document storage using **Amazon S3 with expiring presigned URLs** and enforced **role based access control (RBAC)** across platform endpoints. Automated Java compilation, testing, and containerized deployment to **Amazon ECS** via **Jenkins CI/CD** with **AWS Lambda** for scheduled tasks.
- Contributed to **Agentic AI development** by building autonomous agents, defining modular skills, integrating **MCP tool access**, and deploying **RAG pipelines with guardrails**. Optimized LLM inference costs and response efficiency through **dynamic model routing** across lightweight and advanced LLMs.

PROJECTS

KadalVazhi: Maritime Operations and AI Platform **2026**

Java 25, Spring Boot, Apache Kafka, Python, PostgreSQL, Redis, Offline SQLite Sync *Distributed Microservices*

- Architected a distributed maritime operations platform using **Java 25 and Spring Boot** with an **offline first SQLite WAL sync engine**, ensuring uninterrupted telemetry and catch logging during **7 to 15 day deep sea cellular blackouts** with idempotent cloud reconciliation.
- Built an event driven messaging pipeline with **Apache Kafka partitioned by Harbour ID** for live catch preorders and emergency crew dispatch, integrating **multilingual Whisper and Gemini AI voice advisory** cutting LLM inference costs by **45%** via dynamic model routing.

TicketFlow: Agile Task & AI Voice Workspace Tracker **2026**

React.js, GraphQL, Supabase, Upstash Redis, Node.js, Groq (Llama/Qwen), ElevenLabs, Vercel *Full-Stack Web & AI*

- Engineered a real-time task tracker and Kanban board workspace using **React.js, GraphQL, and Supabase (PostgreSQL)**, deployed on **Vercel** with a lightweight **Node.js micro-service** to eliminate serverless cold starts and optimize query latency.
- Integrated an interactive **ElevenLabs voice assistant** and sub-second LLM inference via **Groq (Llama & Qwen models)**, implementing **Upstash Redis caching** and an automated **model fallback pipeline** for reliable, fault-tolerant issue triage.

TECHNICAL SKILLS

Languages: Java (Core & 21), Python, JavaScript, SQL, C++

Frontend & Backend: React.js, Spring Boot, FastAPI, REST APIs, GraphQL, gRPC, Microservices

Distributed Systems & Cloud: Apache Kafka, Distributed Systems, Redis Caching, Rate Limiting, Docker, Jenkins CI/CD, AWS (EC2, S3, Lambda)

Databases: PostgreSQL, Supabase, DynamoDB, Redis (Upstash), MySQL, SQLite

AI & Systems: Agentic AI, Groq (LLM Inference), ElevenLabs, MCP Tools, Whisper Voice Engine, Gemini API, RAG

ACHIEVEMENTS

- **LeetCode Guardian Badge:** Earned an official **Contest Rating of 2,134** (Top 1.33% worldwide) with **700+ algorithmic problems** solved in **Java and Python** (Graphs, Trees, Dynamic Programming, Sliding Window, Two Pointers, Hash Maps, Linked Lists).
- **Smart India Hackathon (SIH) Team Lead:** Led a 5 member engineering team to build **FarmVista**, an AI precision agriculture platform integrating deep learning computer vision for automated crop disease diagnosis and soil telemetry.

EDUCATION

Knowledge Institute of Technology *Salem, India*

B.Tech in Artificial Intelligence and Data Science | CGPA: 8.1 2021 to 2025

CERTIFICATIONS

Core Java & SQL (EBox) | MySQL Database (IBM) | Machine Learning (Simplilearn) | Deep Learning PyTorch (Guvi)